



KAYA Frame Grabbers Direct GPU Support Guide

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Rev xxx

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1 Introduction

1.1 Safety precautions

Please take the time to read through the precautions listed below in order to prevent preventable and unnecessary injuries and damage to you, other personnel or property. Read these safety instructions carefully prior to your first use of the product, as these precautions contain safety instructions that must be observed. Be sure to follow this manual in order to prevent misuse of product.



Caution! Read Carefully and do not disregard these instructions.

In the event of a failure, disconnect the power supply

Disconnect the power supply immediately and contact our sales personnel for repair. Continuing to use the product in this state may result in a fire or electric shock.

If an unpleasant smell or smoking occurs, disconnect the power supply.

Disconnect the power supply immediately! Continuing to use the product in this state may result in a fire or electric shock. After verifying that no smoking is observed, contact our sales personnel for repair.

Do not disassemble, repair or modify the product.

This may result in a fire or electric shock due to a circuit shortage or heat generation. Contact our sales personnel prior to inspection, modification or repair.

Do not place the product on unstable surfaces.

Otherwise, it may drop or fall, resulting in injury to persons or the camera.

Do not use the product if dropped or damaged.

Otherwise, a fire or electric shock may occur.

Do not touch the product with metallic objects.

Otherwise, a fire or electric shock may occur.

Do not place the product in dusty or humid environments, nor where water may splash.

Otherwise, a fire or electric shock may occur.

Do not wet the product or touch it with wet hands.

Otherwise, the product may fail or it may cause a fire, smoking or electric shock.

Do not touch the gold-plated sections of the connectors on the product.

Otherwise, the surface of the connector may be contaminated by sweat or skin-oil, resulting in contact failure of a connector, malfunction, fire or electric shock due to static electricity discharge.

Do not use or place the product in the following locations.

- Unventilated areas such as closets or bookshelves.
- Near oils, smoke or steam.
- Next to heat sources.
- A closed (and not running) car where the temperature becomes high.
- Static electricity replete locations
- Near water or chemicals.

Otherwise, a fire, electric shock, accident or deformation may occur due to a short circuit or heat generation.

Do not place heavy objects on the product.

Otherwise, the product may be damaged.

Be sure to discharge static electricity from body before touching any sensitive electronic components.

The electronic circuits in your computer and the circuits on the camera are sensitive to static electricity and surges. Improper handling may seriously damage the circuits. In addition, do not let your clothing come in contact with the circuit boards or components. Otherwise, the product may be damaged.

1.2 Disclaimer

This product should only be used for image capturing and processing. **KAYA Vision** assumes no responsibility for any damage that may ensue by using the camera for any purpose other than intended, as previously stated. Without detracting from what was previously written, please be advised that the company takes no responsibility for any damages caused by:

- Earthquake, thunder strike, natural disasters, a fire caused by usage beyond our control, willful and/or accidental misuse and/or use under other abnormal and/or unreasonable conditions
- Secondary damages caused by the use of this product or its unusable state (business interruption or others)
- Use of this product in any manner that contradicts this manual or malfunctions due to connection to other devices.
- Damage to this product that is out of our control or failure due to modification.
- Accidents and/or third parties that may be involved.

Additionally, **KAYA Vision** assumes no responsibility or liability for:

- Erasure or corruption of data caused by the use of this product
- Any consequences or other abnormalities following the use of this product

Repairs to this product are carried out by replacing it on a chargeable basis and not by repairing the faulty device. Non-chargeable replacement is offered for the initial failure, as long as it is reported no later than two weeks post-delivery of the product.

2 Overview

The purpose of this document is to describe the provided functionality of Direct GPU support in KAYA's Frame Grabbers.

Modern GPUs which implement Direct GPU mechanism, allow system to eliminate redundant memory copies thus increasing data integrity and bandwidth by using less resources. The mechanism optimizes the data transfer path by performing as little data copies as possible, subject to GPU's driver implementation, thus reducing transfer latency.

KAYA's Frame Grabbers are able to utilize the Direct GPU capabilities, and perform a data transfer, with as little data copy as the GPU's driver functionality implements.

3 How does it work?

3.1 NVIDIA GPUDirect for Video

NVIDIA GPUs, which implements the “GPUDirect”, eliminate unnecessary memory copies by using shared system memory, which then can be filled by a Frame Grabber. As a result, it reduces CPU and memory usage, as well as data transfer latency from the Frame Grabber to the GPU.

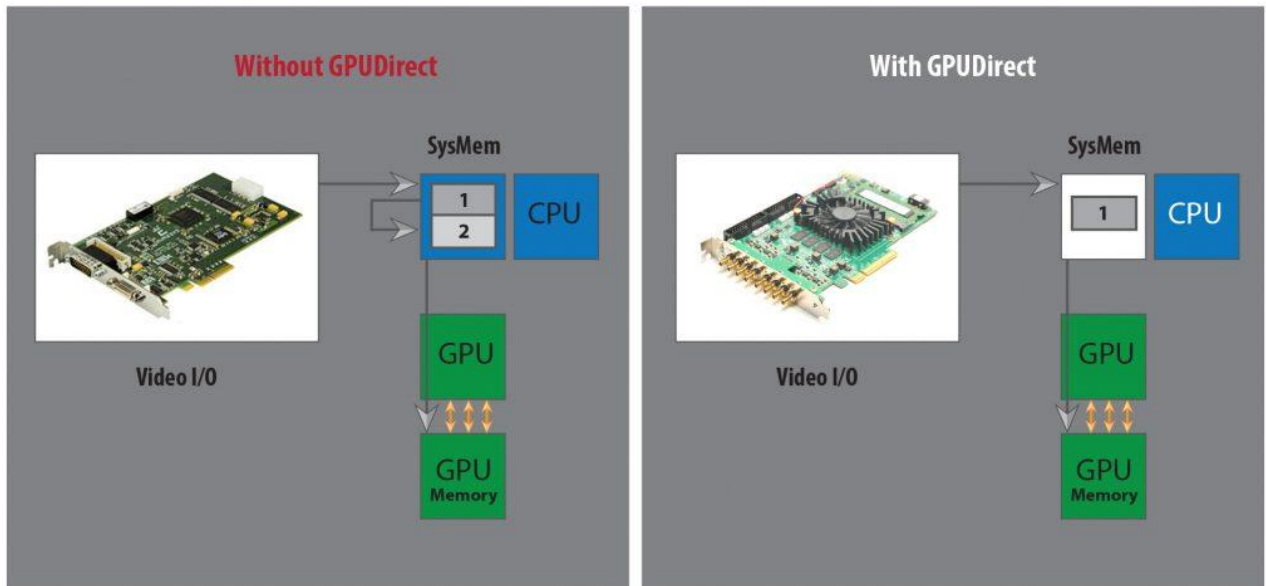


Figure 1 – Direct GPU workflow

More information can be found on NVIDIA’s official site: <https://developer.nvidia.com/gpudirectforvideo>

Each GPU’s SDK provides different functionality, subject to GPU’s driver implementation, regarding Frame Grabber connectivity and data transfer method.

Successful data transfer process from KAYA’s Frame Grabber to destination GPU, differs very little from regular stream creation flow and data transfer. KAYA’s API function sequence should be incorporated in GPU’s native software SDK which will serve as the Host application.

Here are some examples on how to use KAYA’s API with different GPUs:

3.2 API sample demonstrating using NVIDIA GPUDirect for Video with KAYA Vision Point II SDK

This sample demonstrate how to use

Open KYVP_QueuedBuffers_Example, activate macro KYVP_CUDA_BUFFERS and compile and run.

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REFERENCES

Supported vision standards:



Vision Point documentation:



TECHNICAL SUPPORT AND PROFESSIONAL SERVICE

If you searched the documents and could not find the answers you need, contact KAYA Vision support service:

- Create a support request on the web: support.kaya.vision
- Our knowledge base is available on: kb.kaya.vision

Visit us at www.kaya.vision for comprehensive information.

SUBMITTING A SUPPORT REQUEST

When opening a support request, please provide the following information when applicable:

For Frame Grabbers:	For Camera:	For Range Extender:
• Vision Point Diagnostic Info* • Serial number of Frame Grabber • Camera model • SFP+ module model • CoaXPress/Fiber cable model and length • External power or PoCXP • PC motherboard model *In the Vision Point app, use menu option Help > Collect diagnostic info.	• Vision Point Diagnostic Info (or frame grabber being utilized) • Serial Number of Camera • XML dump and/or description of how the camera is being utilized • Description of issue • SFP+ module model • CoaXPress/Fiber cable model and length • External power or PoCXP	• Range Extender Model • Serial Number of Range Extender • SFP+ module model • CoaXPress/Fiber Cable model and length • PC configuration • Operating System • Software Revision • Camera and Frame Grabber Manufacturer and Model



Have questions about pricing, availability, documentation, or custom options?
We're always ready to assist and provide expert guidance.
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